Lab 1: Requirements Engineering and UML Use-Case Modeling
Problem Statement 3: Campus Placement and Internship Pipeline

Student Name: Tanisha Mathur

SRN: PES1UG24CS491

Section: 5-I

1. Requirements Specification Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes in PDF format, extract skill tags, and match student profiles with drive eligibility criteria.	High	Pass: Ineligible applicants are filtered out with clear discrepancy reasons. Fail: An applicant with active backlogs bypasses the filter.	Automates initial screening based on company criteria.
FR-002	Functional	The system shall allow the Placement Officer to set drive eligibility rules such as minimum CGPA, eligible branches, and maximum active backlogs.	High	Pass: Drive is published only when numerical thresholds are valid. Fail: System allows publishing with missing or conflicting criteria.	Lets placement coordinators configure custom drive requirements.

FR-003	Functional	The system shall allow eligible students to apply for active drives and track their real-time application status.	High	Pass: Eligible students submit within the deadline and see status updates. Fail: Ineligible student submits or status is not updated.	Provides transparency for students during ongoing drives.
FR-004	Functional	The system shall schedule shortlisted candidates into dynamic interview slots across multiple rounds.	Medium	Pass: Shortlisted students get interview slots without scheduling conflicts. Fail: Rejected candidate gets a slot or double-booking occurs.	Prevents scheduling clashes during hiring rounds.
FR-005	Functional	The system shall generate digital offer letters and provide accept or reject workflows for selected students.	Medium	Pass: Offer letter is sent to selected candidates with an acceptance deadline. Fail: Offer is generated without coordinator approval.	Manages the final selection and record keeping.
NFR-001	Non-Functional	The system shall encrypt student academic records and offer documents at rest using AES-256.	High	Pass: Database logs confirm all stored files and records are encrypted. Fail: Sensitive data stored in plain text.	Protects student and placement data security.

NFR-002	Non-Functional	The system shall handle up to 2,000 concurrent student applications during peak registration with response time under 1.5 seconds.	High	Pass: Response time stays under 1.5 seconds during peak load tests. Fail: System latency exceeds 1.5 seconds or fails requests.	Prevents portal crashes when drive registrations open.

2. System Actors and Use Cases

Actors:

1. Student Applicant (Primary actor who applies to drives and views status)
2. Placement Officer (Primary actor who sets rules, manages interviews, and issues offers)
3. Document Parser Service (Secondary system actor that extracts text from resumes)

Use Cases:

- UC-01: Apply for Placement Drive
- UC-02: Parse Resume and Extract Skills
- UC-03: Validate Eligibility Criteria
- UC-04: Claim Special Category Relaxation
- UC-05: Manage Interview Queues
- UC-06: Issue Offer Letters
- UC-07: Configure Drive Rules

Key Relationships:

- UC-01 (Apply for Placement Drive) includes UC-03 (Validate Eligibility Criteria).
- UC-03 (Validate Eligibility Criteria) includes UC-02 (Parse Resume and Extract Skills).
- UC-04 (Claim Special Category Relaxation) extends UC-03 (Validate Eligibility Criteria).

3. Use Case Flow Specification

Use Case ID: UC-01 Use Case Name: Apply for Placement Drive Primary Actor: Student Applicant Secondary Actors: Placement Officer, Document Parser Service

Preconditions:

1. The student is registered and logged into the placement portal.
2. An active placement drive is currently open for applications.

3. The student has uploaded a resume in PDF format.

Postconditions:

1. The application status changes to "Submitted" in the portal.
2. The student receives an application confirmation with a reference ID.

Main Success Scenario:

1. Student logs in and navigates to the active drives page.
2. System displays the list of open placement drives with their eligibility criteria.
3. Student selects a drive and clicks on Apply.
4. System invokes the eligibility check (includes Validate Eligibility Criteria).
5. System uses the parser service to read the resume (includes Parse Resume and Extract Skills) and compares student CGPA, branch, and backlog details against drive rules.
6. System verifies that all criteria are satisfied.
7. System shows an application summary and asks for final confirmation.
8. Student clicks Confirm and Submit.
9. System saves the application, assigns an application ID, and updates the status.
10. System displays a success message on the screen.

Alternate Flows:

4a. Ineligible Student:

1. System detects that the student does not meet the minimum CGPA or has active backlogs.
2. System displays an error message stating the exact reason for ineligibility.
3. System prevents submission and returns the student to the dashboard.
4. Use case ends.

5a. Unreadable Resume PDF:

1. System cannot extract text from the uploaded PDF resume.
2. System prompts the student to re-upload a clean, machine-readable PDF resume.
3. Student uploads a valid file.
4. System resumes the flow from Step 5.